

Advanced Natural Language Processing

Measuring Quality in NLP

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Learning Goals Today

- Understand how to measure the quality of NLP algorithms.
 - Experience hands-on the theory of this lecture
-
- **By the end of this lecture you will be able to:**
 - Explain why accuracy is misleading on unbalanced datasets, and compute precision, recall, and $F\beta$ instead
 - Compute perplexity and explain why frontier chat models no longer publish it
 - Compute ROUGE and BLEU scores by hand, including clipping and the brevity penalty
 - Explain why n-gram overlap metrics can be fooled by factually inverted text
 - Match each NLP task (classification, generation, translation, dialogue) to the right evaluation method



Evaluating NLP Tasks



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Teaching note: Every task from here on gets the same treatment: define what “correct” means, then pick the metric that actually measures it.

Why Quantitative Measurements of NLP Algorithms?

Because **we need objective evidence** that an NLP system works as intended.

- **Avoid subjectivity**

Human impressions (“it seems good”) are unreliable.

Quantitative metrics (accuracy, F1, BLEU, perplexity, etc.) give *measurable* performance.

- **Compare algorithms fairly**

Without numbers, we can’t say whether model A is better than model B.

Quantitative evaluation allows benchmarking on shared datasets.

- **Track improvements over time**

NLP development is iterative.

Metrics let us see if a change improves or harms performance.

- **Generalize beyond examples**

A model may look great on a few sample sentences but fail on thousands of unseen cases.

Metrics computed over large datasets measure robustness and generalizability.

- **Make decisions with stakeholders**

In real-world products, decisions need justification:

“We should deploy this model because it reduces error by 12%.”

Ethical Considerations to use Quantitative Measurements

- **Detect Bias & Ensure Fairness**
Measure performance across groups to prevent discrimination.
- **Guarantee Safety & Reliability**
Quantify error rates to avoid harmful failures in sensitive contexts.
- **Enable Accountability & Transparency**
Numbers make model decisions auditable and explainable.
- **Prevent Overconfidence & Misuse**
Evaluate uncertainty so humans know when to intervene.

We need different quality measurements for different NLP tasks

- Quality of classification decisions
- Quality of a language models
- Quality of generated text
- Quality of a translation
- Quality of a dialogue.

Measuring the Quality of Classification Problems in NLP



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Teaching note: Classification quality looks simple (right/wrong) but accuracy alone is misleading whenever the classes are unbalanced — next slides show why.

Examples of Classification Problems in NLP

- Sentence detection
- Tokenization
- POS tagging (detecting syntactic role)
- Subject-Object detection
- Phrase detection
- Co-reference and pronoun resolution
- Negation handling
- Detecting semantic roles (Named Entity Recognition, Sentiment mining, extracting relations, ...)
- Speech recognition

How about Accuracy form the world of machine learning...

- Accuracy = what I do correctly compared relatively to everything I do
- What I do right:
 - Find all the relevant document with a search query = true positive (tp)
 - Do not find all the non- relevant documents with that query = true negative (tn)
- What can I do wrong:
 - Find a number of non-relevant document with a search query = false positive (fp)
 - Do not find a number of relevant documents with that query = false negative (fn)

$$Accuracy = \frac{\# \text{ Correct Results}}{\# \text{ All Possible Results}} = (tp+tn)/(tp+tn+fp+fn)$$

Unbalanced Data Sets

- In Information Retrieval tasks there are often very few (tens to hundreds) relevant documents hidden in many (often millions of billions) irrelevant documents.
- This is called an un-balanced data set.
- Imagine searching the web: you are only interested in a few documents on finding a hotel in Amsterdam out of all websites in the entire world.



High accuracy does not mean you find something useful

- You can build a 99% accurate search engine on a low budget....

A screenshot of a search engine interface. At the top, the text 'snoogle.com' is displayed in a large, stylized font with a blue-to-orange gradient. Below it, the text 'Search for:' is followed by a white rectangular input box. Underneath the input box, the text '0 matching results found.' is displayed in a blue, italicized font. The entire interface is enclosed in a thin black rectangular border.

snoogle.com

Search for:

0 matching results found.

- Caused because in IR there is often a **large imbalance** between relevant documents (**very small**) and non-relevant document (**very large**).
- This is why we need to measure both **precision** and **recall**.

Why not accuracy

- $\text{Accuracy} = (\text{tp} + \text{tn}) / (\text{tp} + \text{fp} + \text{fn} + \text{tn})$

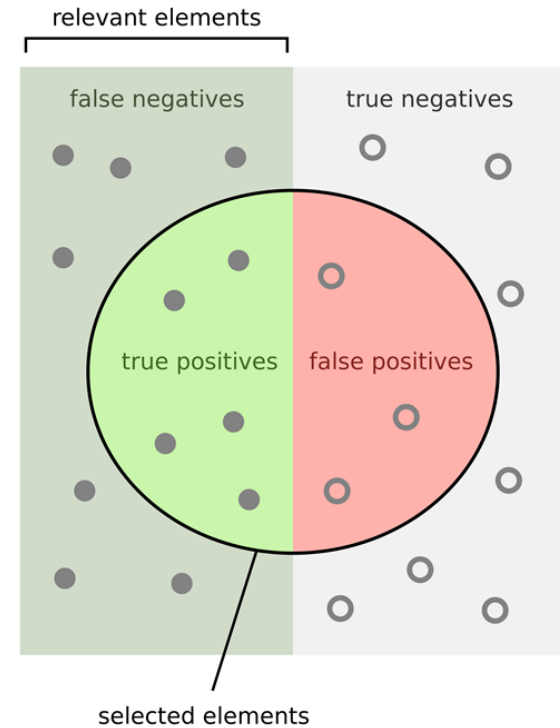
Suppose:

- a set of 100.000 docs in total
- There are 1.000 relevant documents on a certain topic
- My (bad) query finds only 0 relevant = tp, so fn = 1000.
- There are no wrong documents in my result list, so my fp = 0. tn = 99.000
- $\text{Accuracy} = (0 + 99.000) / (0 + 0 + 999 + 99.000) = 99\%$

For a very bad search result!

How about Precision & Recall from the world of information retrieval

- Lack of precision is cost
- Lack of recall is risk



How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Source: https://en.wikipedia.org/wiki/Precision_and_recall

Precision and Recall

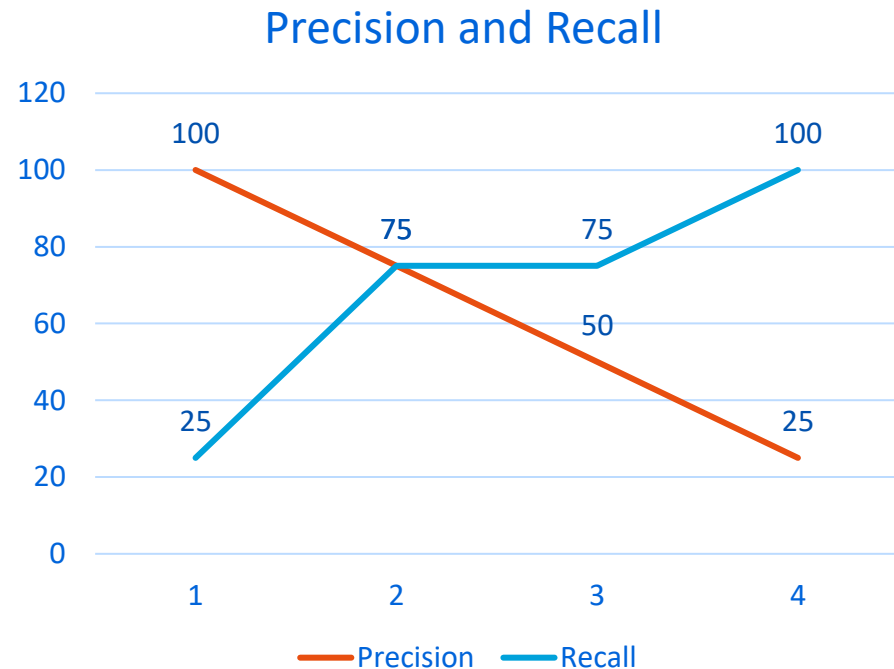
- **Precision:** fraction of retrieved docs that are relevant = $P(\text{relevant} | \text{retrieved})$
- **Recall:** fraction of relevant docs that are retrieved = $P(\text{retrieved} | \text{relevant})$

	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

- Precision $P = \text{tp} / (\text{tp} + \text{fp})$
- Recall $R = \text{tp} / (\text{tp} + \text{fn})$

Precision & recall: reverse proportional

- Precision increasing: AND, W/5, NOT
- Recall increasing: OR, *, ?, Thesaurus, fuzzy.



A combined measure: F

- Combined measure that assesses precision/recall tradeoff is **F measure** (weighted harmonic mean):

$$F = \frac{1}{\alpha \frac{1}{P} + (1-\alpha) \frac{1}{R}} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- People usually use balanced F_1 measure
 - i.e., with $\beta = 1$ or $\alpha = \frac{1}{2}$. $B < 1$ prefer precision, $B > 1$ prefer recall. $B = 1$ and $\alpha = 1/2$, P and R are equally important.
- Harmonic mean is a conservative average
 - See CJ van Rijsbergen, *Information Retrieval*

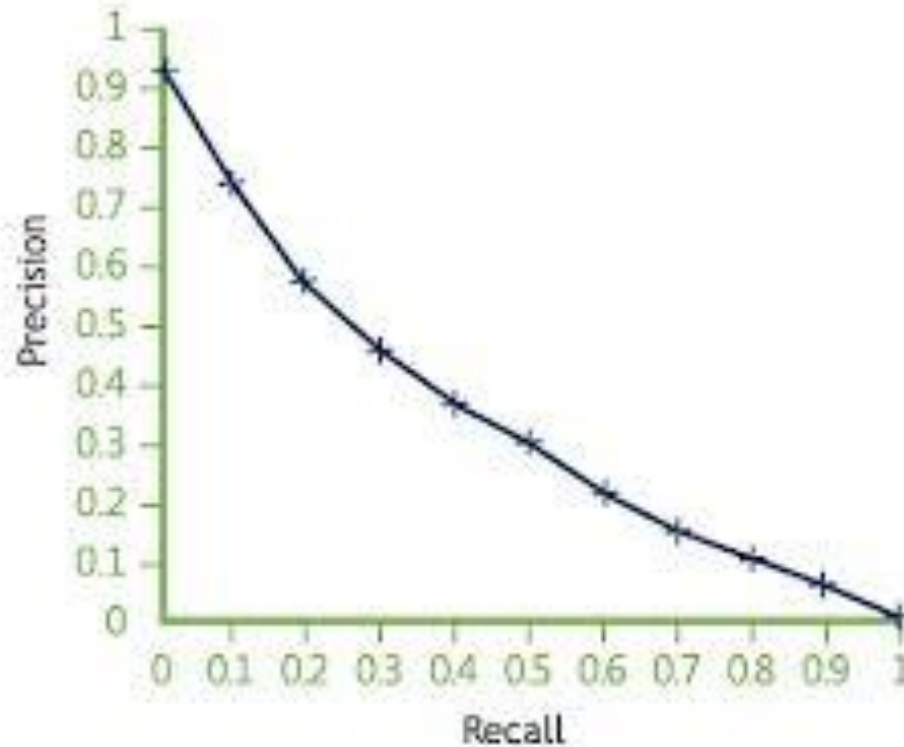
F1 is most popular (balanced P and R)

$$F = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

For $\beta = 1$:

$$F_1 = 2 * \frac{\textit{precision} * \textit{recall}}{\textit{precision} + \textit{recall}}$$

Precision & Recall: 11-points



We use 11-points precision to address this problem:
precision for 11 points of recall.

Measuring the Quality of a Language Model



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Teaching note: A language model has no single “correct answer” to score against — perplexity measures how well it predicts unseen text instead.

Perplexity

- In information theory, perplexity is a measurement of how well a probability distribution or probability model predicts a sample.
- It may be used to compare probability models.
- A low perplexity indicates the probability distribution is good at predicting the sample.

$$PP = 2^{H(p_{LM})}$$

$$H(p_{LM}) = -\frac{1}{n} \log(p_{LM}(w_1, w_2, \dots, w_N))$$

Perplexity

The best language model is one that best predicts an unseen test set

- Gives the highest $P(\text{sentence } W)$

Perplexity is the inverse probability of the test set, normalized by the number of words:

Chain rule:

For bigrams:

$$PP(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 \dots w_N)}}$$

$$PP(W) = \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_i | w_1 \dots w_{i-1})}}$$

$$PP(W) = \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_i | w_{i-1})}}$$

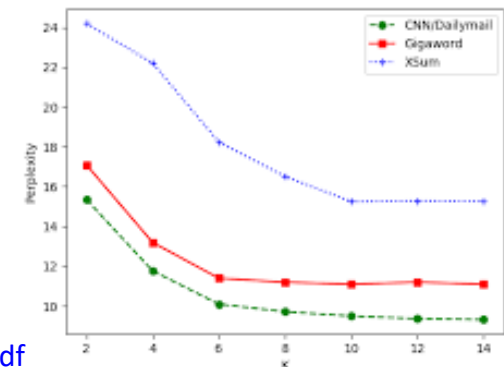
Minimizing perplexity is the same as maximizing probability

Lower perplexity = better model

- Training 38 million words, test 1.5 million words, WSJ

N-gram Order	Unigram	Bigram	Trigram
Perplexity	962	170	109

The perplexity score of GPT-2 calculated on the wikitext dataset was 16.4541. The perplexity of LLaMA-2 7B calculated on its diverse test dataset is between 5.0 - 6.0.



Is Perplexity still a useful measurement?

- GPT-2 and LLaMA-2 publish perplexity because they are *pretrained language models*.
- GPT-5, Gemini 2, Claude 3, etc., do **not** publish perplexity because they are *instruction-tuned chat models*.
- If measured, GPT-5 / Gemini would likely score **below 3** — significantly better than LLaMA-2 and GPT-2.



Modern frontier models (GPT-4, GPT-5, Gemini 1.5/2.0, Claude 3, etc.) do not publish perplexity for three reasons:

1. RLHF / instruction tuning breaks perplexity comparability. Perplexity is defined for next-token prediction on a fixed language-model objective. Once a model is instruction-tuned (SFT + RLHF), it is no longer a pure next-token model, so perplexity loses meaning across models.
2. Companies benchmark on tasks instead (MMLU, GSM8K, HumanEval, etc.)
 - GPT-5 reports improvements on accuracy and hallucination rates, not perplexity.
 - Google Gemini and Anthropic Claude do the same.

Quality of Summaries



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Teaching note: Unlike classification, a summary can be “close enough” in many different ways — ROUGE measures overlap with a reference, not correctness.

Some Background on Terminology Used

- 1-gram = single character or token
- 2-gram or bigram = 2 characters or token
- 3-gram = 3 characters or token
- N-gram = n characters or token
- Skip-bigram = any combination of 2 in a larger text

Skip Bigrams



Source Text	Training Samples					
<table><tr><td>The</td><td>quick</td><td>brown</td></tr></table> fox jumps over the lazy dog. ➡	The	quick	brown	(the, quick) (the, brown)		
The	quick	brown				
The <table><tr><td>quick</td><td>brown</td><td>fox</td></tr></table> jumps over the lazy dog. ➡	quick	brown	fox	(quick, the) (quick, brown) (quick, fox)		
quick	brown	fox				
The quick <table><tr><td>brown</td><td>fox</td><td>jumps</td></tr></table> over the lazy dog. ➡	brown	fox	jumps	(brown, the) (brown, quick) (brown, fox) (brown, jumps)		
brown	fox	jumps				
The <table><tr><td>quick</td><td>brown</td><td>fox</td><td>jumps</td><td>over</td></tr></table> the lazy dog. ➡	quick	brown	fox	jumps	over	(fox, quick) (fox, brown) (fox, jumps) (fox, over)
quick	brown	fox	jumps	over		

How good is my summary?

- ROUGE-N: overlap of n-grams between candidate (C) and reference (R) summary:
 - ROUGE-1: 1-grams
 - ROUGE-2: bigrams
 - ROUGE-L: longest common subsequence (LCS)

Other variants:

- ROUGE-W: Weighted LCS-based statistics that favor consecutive LCSes.
- ROUGE-S: Skip-bigram based co-occurrence statistics. Skip-bigram is any pair of words in their sentence order.
- ROUGE-SU: Skip-bigram plus unigram-based co-occurrence statistics.

For each we can calculate precision, recall and F-1:

Rouge Precision = ratio of the number of unigrams in Candidate text that also appear also in Reference text

Rouge Recall = ratio of the number of unigrams in Reference that appear also in Candidate

Rouge F-1 = $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$



Example

Reference: The cat is on the mat.

Candidate: The cat and the dog.

- ROUGE-1 Precision = $3/5 = 0.6$
- ROUGE-1 Recall = $3/6 = 0.5$
- ROUGE-1 F1-score = $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall}) = 2 * 0,3 / 1,1 = 0.45$

Example-2

Reference: The cat is on the mat.

Candidate: The cat and the dog.

- ROUGE-2 precision is the ratio of the number of 2-grams in C that appear also in R (only the 2-gram “the cat”), over the number of 2-grams in $C = 1/5$
- ROUGE-1 recall is the ratio of the number of 2-grams in R that appear also in C (only the 2-gram “the cat”), over the number of 2-grams in $R = 1/6$
- F1 follows as before ...

Example-3

Reference: The cat is on the mat.

Candidate: The cat and the dog.

- The LCS is the 3-gram “the cat the” (remember that the words are not necessarily consecutive), which appears in both R and C.
- ROUGE-L precision is the ratio of the length of the LCS, over the number of unigrams in C = $3/5$
- ROUGE-L recall is the ratio of the length of the LCS, over the number of unigrams in R = $3/6$
- F1 follows as before...

The CNN/Daily Mail Dataset

- CNN/Daily Mail is a dataset for text summarization. Human generated abstractive summary bullets were generated from news stories in CNN and Daily Mail websites as questions (with one of the entities hidden), and stories as the corresponding passages from which the system is expected to answer the fill-in-the-blank question. The authors released the scripts that crawl, extract and generate pairs of passages and questions from these websites.
- In all, the corpus has 286,817 training pairs, 13,368 validation pairs and 11,487 test pairs, as defined by their scripts. The source documents in the training set have 766 words spanning 29.74 sentences on an average while the summaries consist of 53 words and 3.72 sentences.

See:

<https://www.kaggle.com/datasets/gowrishankarp/news-paper-text-summarization-cnn-dailymail>

Rouge	metrics	CNN/Daily Mail dataset
Rouge-1	precision	0.35
	recall	0.41
	f-score	0.38
Rouge-2	precision	0.33
	recall	0.36
	f-score	0.34
Rouge-L	precision	0.32
	recall	0.4
	f-score	0.38

Rouge can be high, but depending on what is left out or added, the quality can really differ!

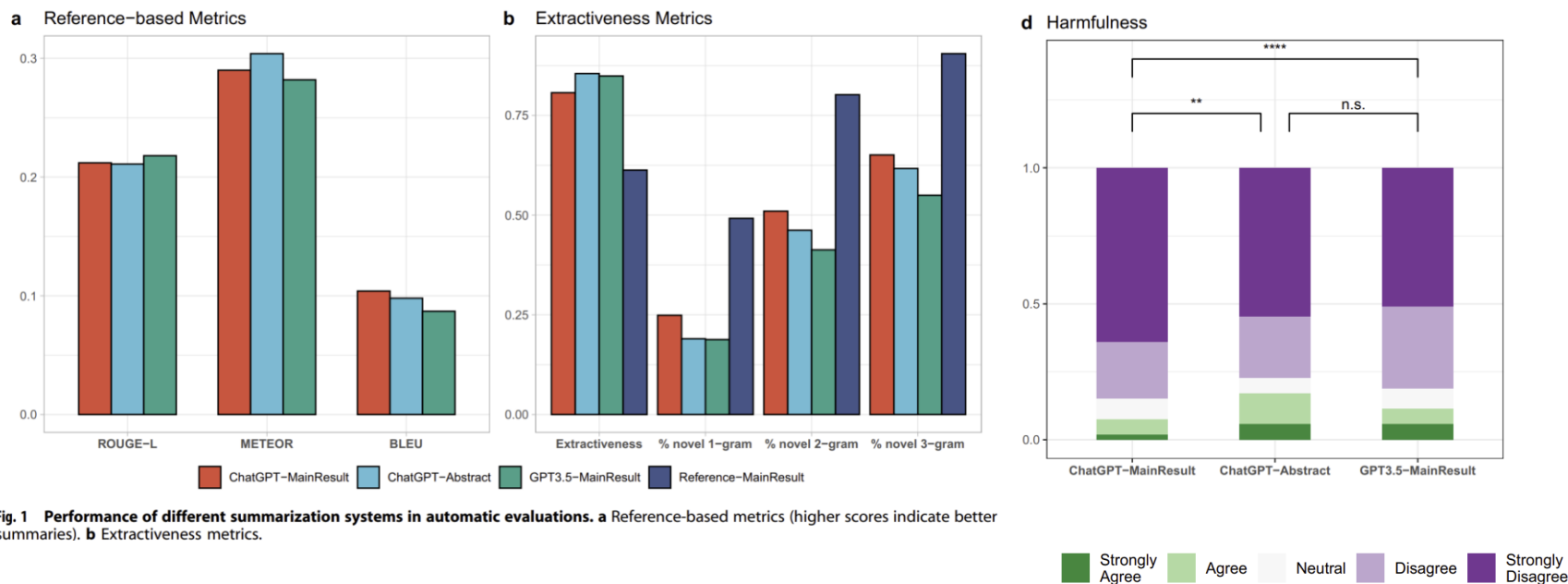


Fig. 1 Performance of different summarization systems in automatic evaluations. **a** Reference-based metrics (higher scores indicate better summaries). **b** Extractiveness metrics.

“Automatic metrics such as ROUGE and BLEU may provide misleading confidence in LLM outputs, because they do not capture factual consistency, coverage of salient information, or potential for harm—especially in specialized/high-stakes domains.” In short: good scores on ROUGE/BLEU do not guarantee safe, trustworthy summaries.

Measuring the Quality of Machine Translation



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Teaching note: Translation has the same problem as summarization, amplified: many correct translations exist for one source sentence.

The Challenge of Measuring Quality for Summaries and Translations...

There is no single “correct” translation.

In natural language, the same meaning can be expressed in many valid ways:

- different word order
- different synonyms
- implicit vs. explicit phrasing
- stylistic and cultural choices

Translation quality is hard to measure because meaning can be correct even when the words don't match.

BLEU

- The Bilingual Evaluation Understudy Score, or BLEU for short, is a metric for evaluating a generated sentence to a reference sentence.
- A perfect match results in a score of 1.0, whereas a perfect mismatch results in a score of 0.0.

m is number of words from C found in R
 w_t is total number of words in R

Example of poor machine translation output with high precision

Candidate	the	the	the	the	the	the	the
Reference 1	the	cat	is	on	the	mat	
Reference 2	there	is	a	cat	on	the	mat

Of the seven words in the candidate translation, all of them appear in the reference translations. Thus the candidate text is given a unigram precision of,

$$P = \frac{m}{w_t} = \frac{7}{7} = 1$$

Comparing metrics for candidate "the the cat"

Model	Set of grams	Score
Unigram	"the", "the", "cat"	$\frac{1 + 1 + 1}{3} = 1$
Grouped Unigram	"the"*2, "cat"*1	$\frac{1 + 1}{2 + 1} = \frac{2}{3}$
Bigram	"the the", "the cat"	$\frac{0 + 1}{2} = \frac{1}{2}$

BLEU

- BLEU's output is always a number between 0 and 1. This value indicates how similar the candidate text is to the reference texts, with values closer to 1 representing more similar texts.
- Few human translations will attain a score of 1, since this would indicate that the candidate is identical to one of the reference translations.

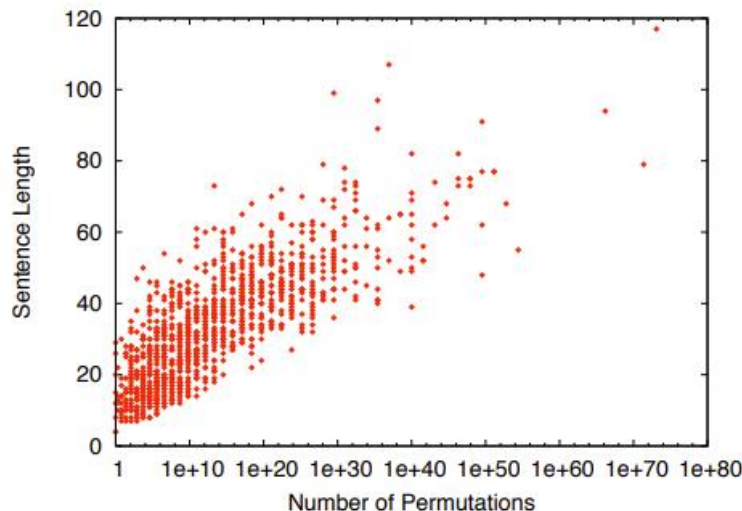


Figure 1: Scatterplot of the length of each translation against its number of possible permutations due to bigram mismatches for an entry in the 2005 NIST MT Eval

See also: Papineni, Kishore; Roukos, Salim; Ward, Todd; Zhu, Wei-Jing (2001). "[BLEU](#)". *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics - ACL '02*. Morristown, NJ, USA: Association for Computational Linguistics. doi:[10.3115/1073083.1073135](#)

• Callison-Burch, C., Osborne, M. and Koehn, P. (2006) "[Re-evaluating the Role of BLEU in Machine Translation Research](#)" in *11th Conference of the European Chapter of the Association for Computational Linguistics: EACL 2006* pp. 249–256

• Doddington, G. (2002) "[Automatic evaluation of machine translation quality using n-gram cooccurrence statistics](#)" in *Proceedings of the Human Language Technology Conference (HLT), San Diego, CA* pp. 128–132

• Coughlin, D. (2003) "[Correlating Automated and Human Assessments of Machine Translation Quality](#)" in *MT Summit IX, New Orleans, USA* pp. 23–27

BLEU versus Human Assessment

- 4 = **Ideal**: Not necessarily a perfect translation, but grammatically correct, and with all information accurately transferred.
- 3 = **Acceptable**: Not perfect (stylistically or grammatically odd), but definitely comprehensible, AND with accurate transfer of all important information.
- 2 = **Possibly Acceptable**: Possibly comprehensible (given enough context and/or time to work it out); some information transferred accurately.
- 1 = **Unacceptable**: Absolutely not comprehensible and/or little or no information transferred accurately.

Human Assessment



Acceptable Translation = 0.35-0.45
BLEU

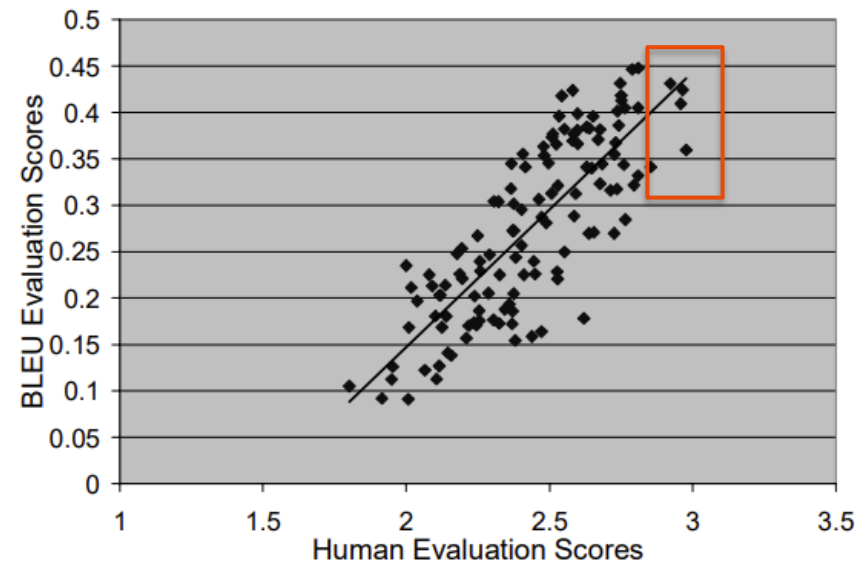
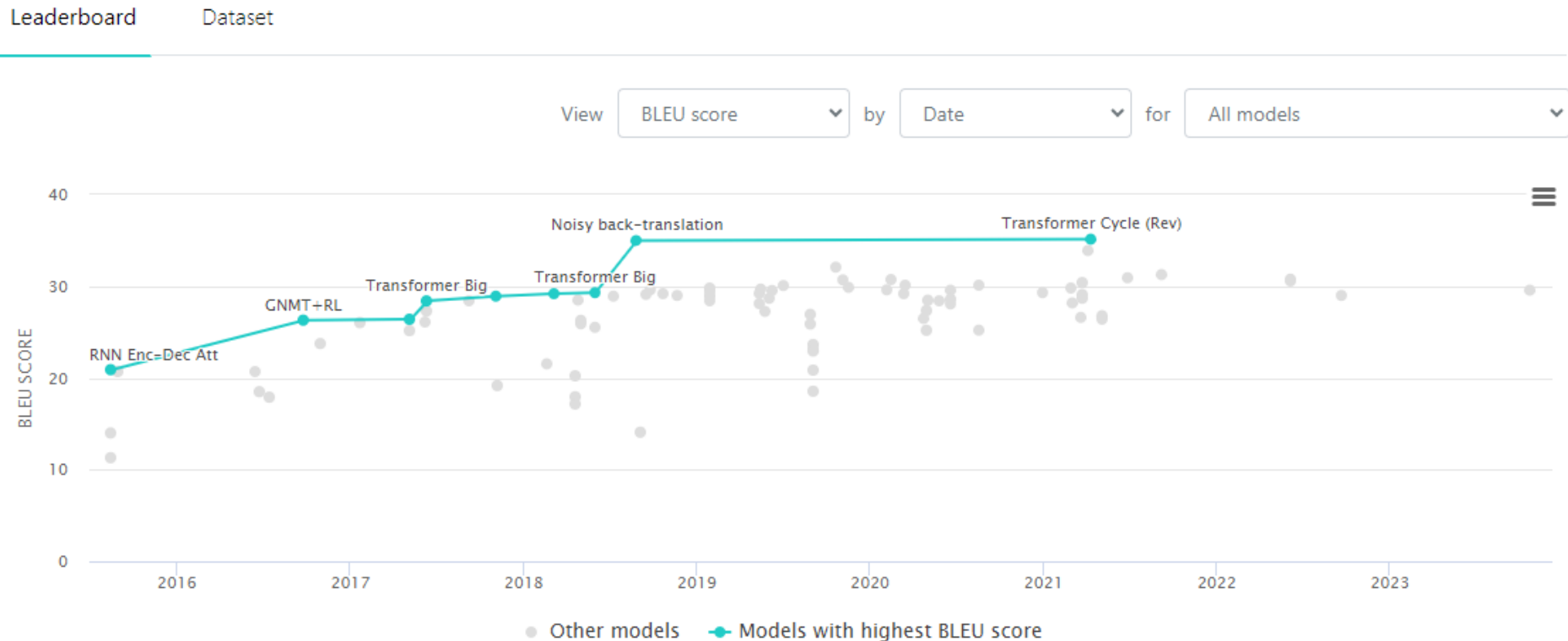


Figure 1: BLEU vs. Human; 124 250-400 sentence evaluations; 7 language pairs; 6 MT systems.

SOTA Machine Translation

Machine Translation on WMT2014 English-German



BLEU

- Count number of equal sentence
 - Example:
 - Reference: “Back to the Future” premiered 30 years ago
 - MT: “Back to the Future” had premiered 30 years ago
- Idea:
 - N-gram overlap

Teaching note: Counting exact sentence matches is almost always 0 in practice — that's exactly why BLEU instead counts overlapping n-grams (next slides).

Example

- Reference: “Back to the Future” premiered 30 years ago
- MT: “Back to the Future” had premiered 30 years ago
- 1-gram: 8/9
- 2-gram: 6/8
- 3-gram: 4/7
- 4-gram: 2/6

Geometric Mean Formula

$$\bar{X}_{geom} = \sqrt[n]{\prod_{i=1}^n x_i} = \sqrt[n]{x_1 \cdot x_1 \cdot \dots \cdot x_n}$$

- Geometric Mean: $\sqrt[4]{\frac{8}{9} * \frac{6}{8} * \frac{4}{7} * \frac{2}{6}}$

Short Sentences

- Reference: “Back to the Future” premiered 30 years ago
- MT C: “Back to the Future”
- 1-gram:
- 2-gram:
- 3-gram:
- 4-gram:

Short Sentences: Brevity Penalty

- Reference: “Back to the Future” premiered 30 years ago
- MT C: “Back to the Future”

- 1-gram:4/4
- 2-gram:3/3
- 3-gram:2/1
- 4-gram:1/1

r- count of words in a reference translation R
c- count of words in a candidate translation C

$$BP = \begin{cases} 1 & , c > r \\ e^{1-r/c} & , c \leq r \end{cases}$$

$$BLEU = \sqrt[4]{P_1 * P_2 * P_3 * P_4 * BP}$$

- Brevity Penalty: $1 - 8/4 = 0.5 \rightarrow BP = 0.77$

We add brevity penalty to handle too short translations. Brevity Penalty(BP) will be 1.0 when the candidate translation length is the same as any reference translation length. The closest reference sentence length is the “best match length.”

Bleu: Clipping – avoid over-generalization of common words such as “the”

- Reference 1: the new movie
- Reference 2: the new film
- MT: the the

- $P_1 = 2/2$

$$p_n = \frac{\sum_{C \in \{\text{Candidates}\}} \sum_{n\text{-gram} \in C} \text{Count}_{\text{clip}}(n\text{-gram})}{\sum_{C' \in \{\text{Candidates}\}} \sum_{n\text{-gram}' \in C'} \text{Count}(n\text{-gram}')}$$

- Clipping:
 - Max count of n-gram in any reference.
 - Clipped count: We clip (remove) the matching count for an n-gram by the largest count observed in any single reference.
 - Avoid over-generalizing matching words (“the”, “new”).



Alternative to BLEU = METEOR

METEOR (Metric for Evaluation of Translation with Explicit ORdering) is a metric for the evaluation of machine translation output.

The metric is based on the harmonic mean of unigram precision (P) and recall (R), with recall weighted higher than precision.

It also has several features that are not found in other metrics, such as stemming and synonymy matching, along with the standard exact word matching.

Designed to fix some of the problems found in the more popular BLEU metric, and also produce good correlation with human judgement at the sentence or segment level.

It also has a penalty (p) mechanism



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the cat sat on the mat
on the mat sat the cat



the cat sat on the mat
on the mat sat the cat



$$P = \frac{m}{w_t} \quad R = \frac{m}{w_r} \quad F_{mean} = \frac{10PR}{R + 9P}$$

$$p = 0.5 \left(\frac{c}{u_m} \right)^3 \quad M = F_{mean} (1 - p)$$

You can find a few examples on:

<https://en.wikipedia.org/wiki/METEOR>

Another Alternative: COMET

- COMET (Crosslingual Optimized Metric for Evaluation of Translation) is a metric for automatic evaluation of machine translation that calculates the similarity between a machine translation output and a reference translation using token or sentence embeddings.
- It is based on similarity of vector representations.

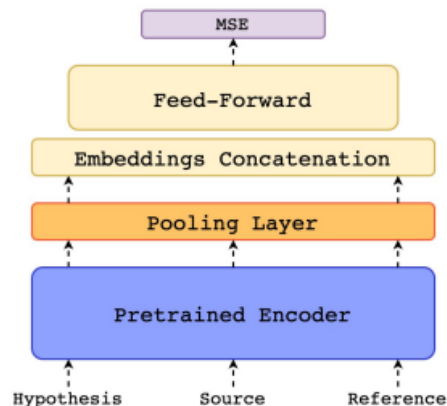


Figure 1: Estimator model architecture. The source, hypothesis and reference are independently encoded using a pretrained cross-lingual encoder. The resulting word embeddings are then passed through a pooling layer to create a sentence embedding for each segment. Finally, the resulting sentence embeddings are combined and concatenated into one single vector that is passed to a feed-forward regressor. The entire model is trained by minimizing the Mean Squared Error (MSE).

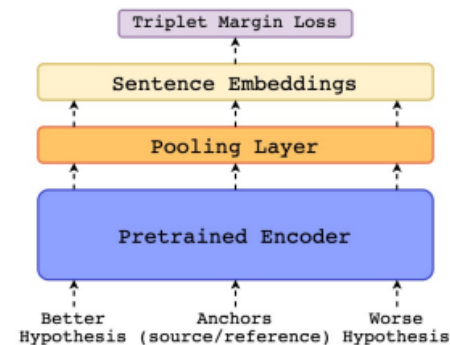
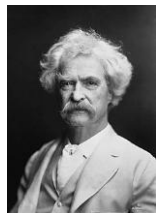


Figure 2: Translation Ranking model architecture. This architecture receives 4 segments: the source, the reference, a “better” hypothesis, and a “worse” one. These segments are independently encoded using a pre-trained cross-lingual encoder and a pooling layer on top. Finally, using the triplet margin loss (Schroff et al., 2015) we optimize the resulting embedding space to minimize the distance between the “better” hypothesis and the “anchors” (source and reference).

Metrics such as BLEU and METEOR struggle to accurately correlate with human judgement at segment level and fail to adequately differentiate the highest performing MT systems

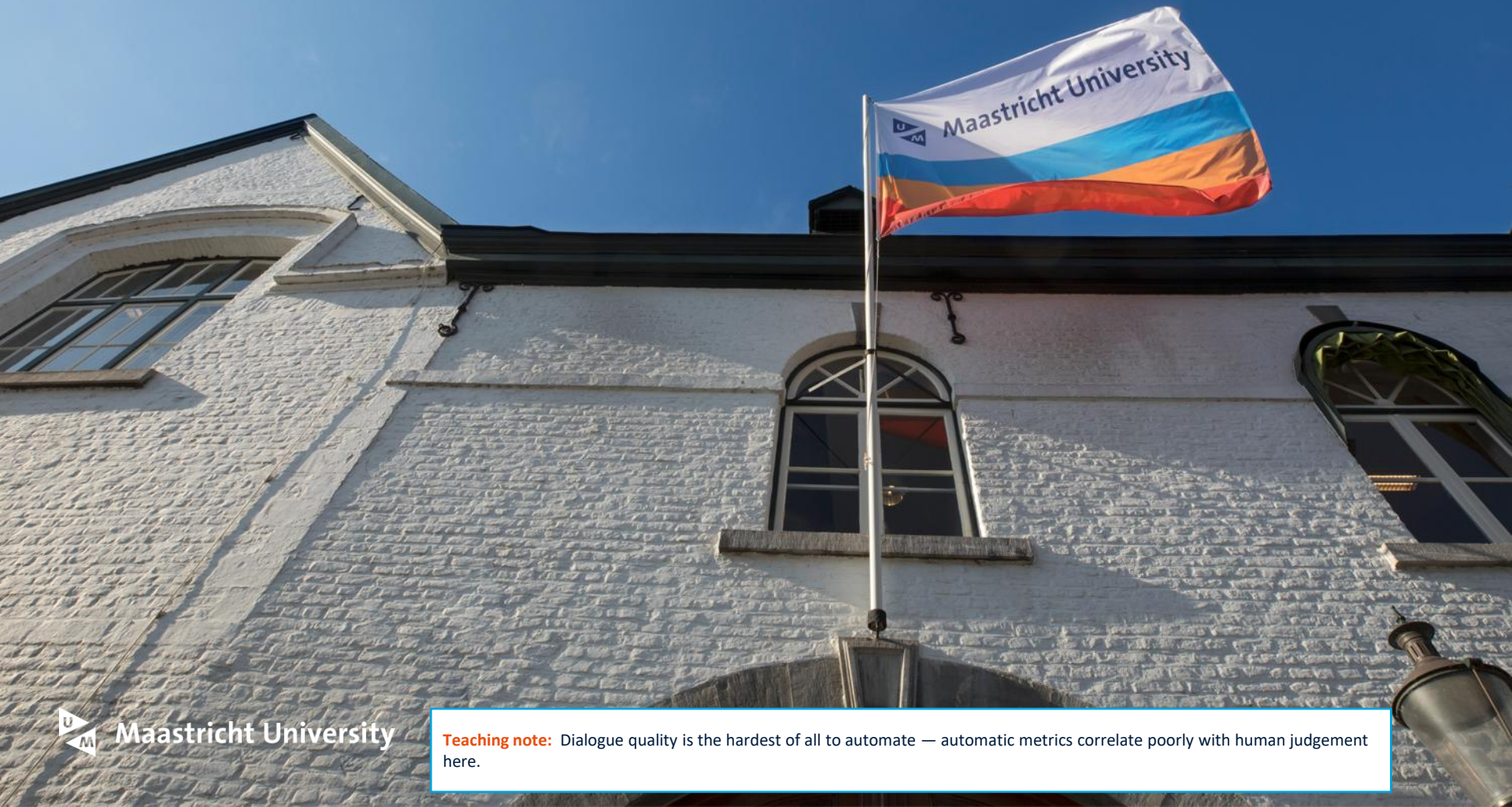
More Alternatives for BLEU ...

- chrF: CHaRacter-level F-score. chrF (CHaRacter-level F-score) is a metric for machine translation evaluation that calculates the similarity between a machine translation output and a reference translation using character n-grams, not word n-grams (<https://machinetranslate.org/chrF>).
- YiSi: YiSi is a metric for automatic evaluation of machine translation that calculates the similarity between a machine translation output and a reference translation using sentence representation. The term YiSi comes from the Cantonese word 意思, which means 'meaning' (<https://machinetranslate.org/yisi>).
- BERTScore: BERTScore is a metric for automatic evaluation of machine translation that calculates the similarity between a machine translation output and a reference translation using sentence representation (<https://machinetranslate.org/bertscore>).
- BLEURT: a learned evaluation metric based on BERT that can model human judgments with a few thousand possibly biased training examples. (<https://arxiv.org/pdf/2004.04696.pdf>).
- TER: Translation Error Rate (TER) is a metric for automatic evaluation of machine translation that calculates the minimum number of edits (normalized by the average length of the references) required to change a machine translation output into one of the references (<https://machinetranslate.org/ter>).



"Lies, damned lies, and statistics"

Quality of a Q&A / Dialog system



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Teaching note: Dialogue quality is the hardest of all to automate — automatic metrics correlate poorly with human judgement here.

Chatbots are evaluated by humans

Automatic evaluations (BLEU/ROUGE, dot products) are generally not used for chatbots. They correlate poorly with human judgements.

- **Participant evaluation:** The human who talked to the chatbot assigns a score
- **Observer evaluation:** third party who reads a transcript of a human/chatbot conversation assigns a score.

Participant evaluation of See et al. (2019)

Human chats with model for 6 turns and rates 8 dimensions of quality: avoiding repetition, interestingness, making sense, fluency, listening, inquisitiveness, humanness, engagingness, e.g.:

(1) Avoiding Repetition: How repetitive was this user?

Repeated themselves over and over • Sometimes said the same thing twice • Always said something new

(2) Making sense: How often did this user say something which didn't make sense?

Never made any sense • Most responses didn't make sense • Some responses didn't make sense • Everything made perfect sense

(3) Engagingness: How much did you enjoy talking to this user?

Not at all • A little • Somewhat • A lot

Observer evaluation: acute-eval

Annotators look at two conversations (A + B) and decide which one is better:

- **Engagingness:** Who would you prefer to talk to for a long conversation?
- **Interestingness:** If you had to say one of these speakers is interesting and one is boring, who would you say is more interesting?
- **Humanness:** Which speaker sounds more human?
- **Knowledgeable:** If you had to say that one speaker is more knowledgeable and one is more ignorant, who is more knowledgeable?

Task completion success

1. Slot Error Rate for a Sentence

$$\frac{\text{\# of inserted/deleted/substituted slots}}{\text{\# of total reference slots for sentence}}$$

2. End-to-end evaluation (Task Success) average of all slots

Teaching note: Worked example with real slot values (PERSON/TIME/ROOM) follows on the next slide.

Evaluation Metrics: Slot error rate

“Make an appointment with Chris at 10:30 in Gates 104”

Slot	Filler
PERSON	Chris
TIME	11:30 a.m.
ROOM	Gates 104

Slot error rate: 1/3

Task success: At end, was the correct meeting added to the calendar?

More fine-grained metrics: User Satisfaction Survey

TTS = Text-to-Speech Synthesis or Speech Generation
ASR = Automatic Speech Recognition

TTS Performance	Was the system easy to understand ?
ASR Performance	Did the system understand what you said?
Task Ease	Was it easy to find the message/flight/train you wanted?
Interaction Pace	Was the pace of interaction with the system appropriate?
User Expertise	Did you know what you could say at each point?
System Response	How often was the system sluggish and slow to reply to you?
Expected Behavior	Did the system work the way you expected it to?
Future Use	Do you think you'd use the system in the future?

Figure 24.20 User satisfaction survey, adapted from [Walker et al. \(2001\)](#).

Walker, M. A., C. A. Kamm, and D. J. Litman. 2001. Towards developing general models of usability with PARADISE. *Natural Language Engineering: Special Issue on Best Practice in Spoken Dialogue Systems*, 6(3):363–377.

Stanford Question Answering Dataset (SQuAD)

Answer type	Percentage	Example
Date	8.9%	19 October 1512
Other Numeric	10.9%	12
Person	12.9%	Thomas Coke
Location	4.4%	Germany
Other Entity	15.3%	ABC Sports
Common Noun Phrase	31.8%	property damage
Adjective Phrase	3.9%	second-largest
Verb Phrase	5.5%	returned to Earth
Clause	3.7%	to avoid trivialization
Other	2.7%	quietly

Table 2: We automatically partition our answers into the following categories. Our dataset consists of large number of answers beyond proper noun entities.

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **grau-pel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called “showers”.

What causes precipitation to fall?

gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

grau-pel

Where do water droplets collide with ice crystals to form precipitation?

within a cloud

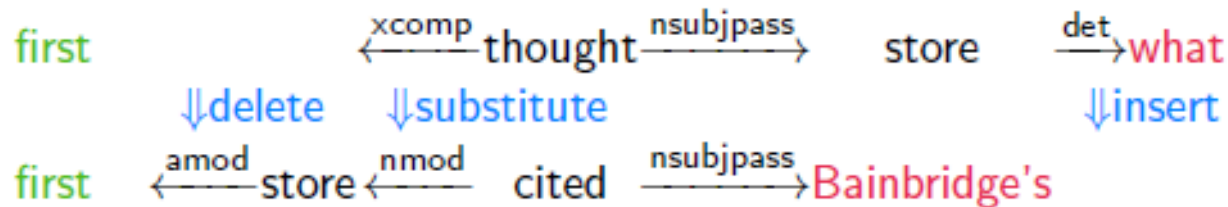
Figure 1: Question-answer pairs for a sample passage in the SQuAD dataset. Each of the answers is a segment of text from the passage.

SQuAD: Dealing with Syntactic Divergence

Q: **What** department store is thought to be the **first** in the world?

S: **Bainbridge's** is often cited as the world's **first** department store.

Path:



Edit cost:

1 +2 +1=4

Figure 3: An example walking through the computation of the syntactic divergence between the question Q and answer sentence S.

A Really Good Benchmark to Evaluate
Generated Text by GPT for Certain Business
Application.

Edit Distance (aka Levenshtein distance)

- In information theory, linguistics, and computer science, the Levenshtein distance is a string metric for measuring the difference between two sequences.
- Informally, the Levenshtein distance between two words is the minimum number of single-character edits (insertions, deletions or substitutions) required to change one word into the other.
- It is named after the Soviet mathematician Vladimir Levenshtein, who considered this distance in 1965.



The Levenshtein distance between two strings a , b (of length $|a|$ and $|b|$ respectively) is given by $\text{lev}(a, b)$ where

$$\text{lev}(a, b) = \begin{cases} |a| & \text{if } |b| = 0, \\ |b| & \text{if } |a| = 0, \\ \text{lev}(\text{tail}(a), \text{tail}(b)) & \text{if } a[0] = b[0], \\ 1 + \min \begin{cases} \text{lev}(\text{tail}(a), b) \\ \text{lev}(a, \text{tail}(b)) \\ \text{lev}(\text{tail}(a), \text{tail}(b)) \end{cases} & \text{otherwise,} \end{cases}$$

where the tail of some string x is a string of all but the first character of x , and $x[n]$ is the n th character of the string x , counting from 0.

Note that the first element in the minimum corresponds to deletion (from a to b), the second to insertion and the third to replacement.

Example Edit Distance

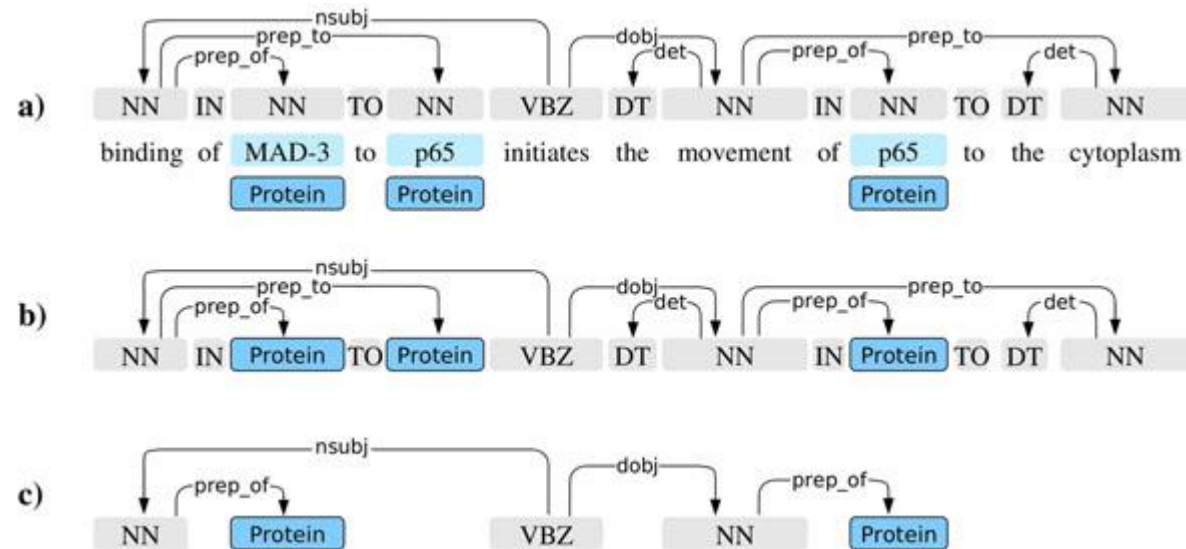
For example, the Levenshtein distance between "kitten" and "sitting" is 3, since the following 3 edits change one into the other, and there is no way to do it with fewer than 3 edits:

1. **k**itten → **s**itten (substitution of "s" for "k"),
2. **sitten** → sitt**i**n (substitution of "i" for "e"),
3. sittin → sitt**ing** (insertion of "g" at the end).

We can use the edit distance in to measure similarity between answers in Q&A tasks.

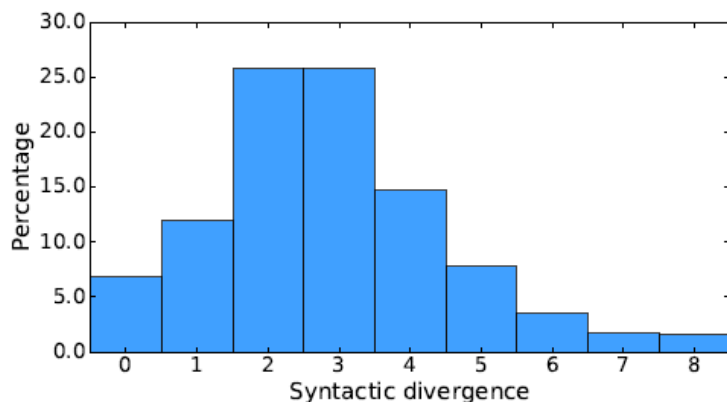
Comparing Edit distance on Character or Word edits would not work

- So, we look at the dependency structures without the actual words from the lexicon:



Unlexicalized shortest path representation a) Applied annotations with original sentence text. b) Unlexicalized representation. c) Shortest path connecting two gene/protein mentions.

SQuAD Measuring Q&A

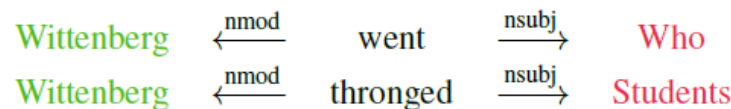


(a) Histogram of syntactic divergence.

Q: Who went to Wittenberg to hear Luther speak?

S: Students thronged to Wittenberg to hear Luther speak.

Path:

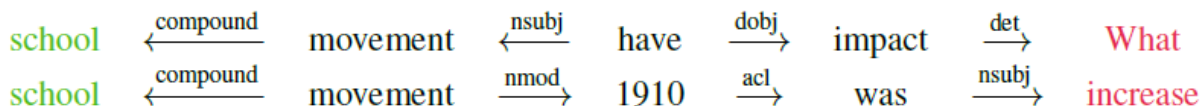


(b) An example of a question-answer pair with edit distance 0 between the dependency paths (note that lexical variation is ignored in the computation of edit distance).

Q: What impact did the high school education movement have on the presence of skilled workers?

S: During the mass high school education movement from 1910 – 1940 , there was an increase in skilled workers.

Path:



(c) An example of a question-answer pair with edit distance 6.

Figure 4: We use the edit distance between the unlexicalized dependency paths in the question and the sentence containing the answer to measure *syntactic divergence*.

Break



Benchmarks for Language Understanding

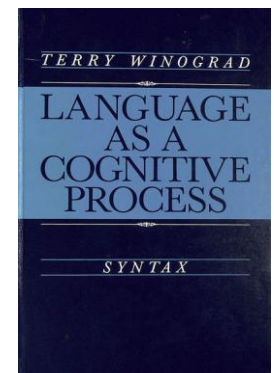


Maastricht University

Teaching note: These benchmarks test understanding and reasoning directly, rather than any single downstream task.

Winograd schema challenge

- The Winograd schema challenge (WSC) is a test of machine intelligence proposed by Hector Levesque, a computer scientist at the University of Toronto. Designed to be an improvement on the Turing test, it is a multiple-choice test that employs questions of a very specific structure: they are instances of what are called Winograd schemas, named after Terry Winograd, professor of computer science at Stanford University.
- On the surface, Winograd schema questions simply require the resolution of anaphora: the machine must identify the antecedent of an ambiguous pronoun in a statement. This makes it a task of natural language processing, but Levesque argues that for Winograd schemas, the task requires the use of knowledge and commonsense reasoning.



Winograd Scheme Example

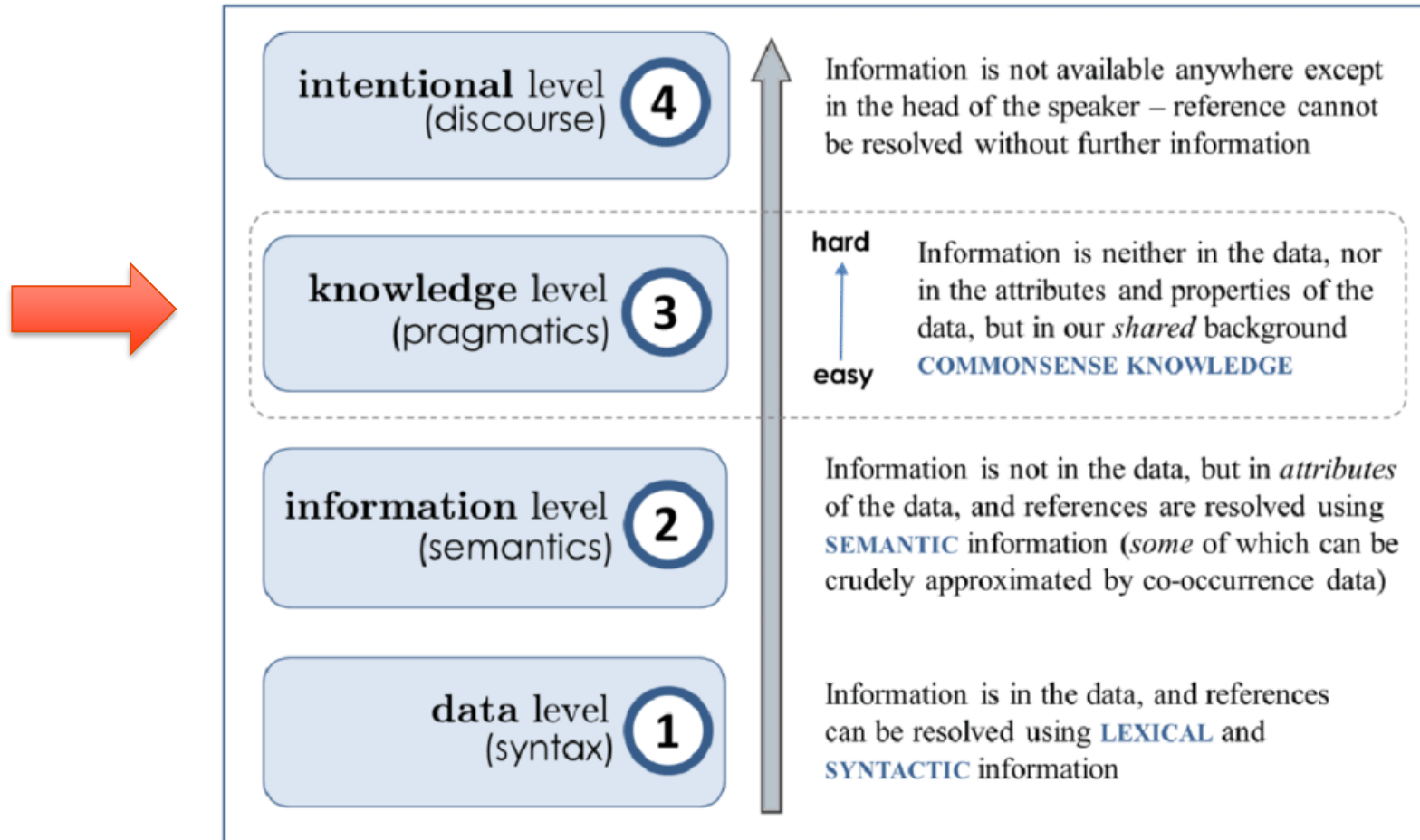
“Dave told everyone in school that he wants to be a guitarist, because he thinks it is a great sounding instrument.”

Short of accessing knowledge structures that somehow encode the commonsense facts:

- that a-guitarist-plays-a-guitar, and
- that a-guitar-is-a-musical-instrument,

How to resolve “it” in since the correct referent is not even in the data, but is an object that is implicitly assumed by the semantic and cognitive content of the text.

Winograd Scheme in the data-information-knowledge continuum



Winograd Scheme Examples

Typical Winograd schemas example

Co-reference cannot be resolved without commonsense knowledge.

- Statement: **Marry** made sure to thank **Susan** for all the help *she* had received.
- Q: who had received the help?
- Answer: Marry

Commonsense knowledge: **receive help** → **thank**

(a) Is correct
(b) Is in-correct

-
- I(a) **The city councilmen** refused the demonstrators a permit because *they* feared violence.
- I(b) The city councilmen refused **the demonstrators** a permit because *they* advocated violence.
-
- II(a) James asked **Robert** for a favor, but *he* refused.
- II(b) **James** asked Robert for a favor, but *he* was refused.
-
- III(a) **Keith** fired Blaine but *he* did not regret.
- III(b) Keith fired **Blaine** although *he* is diligent.
-
- IV(a) Emma did not pass the ball to **Janie**, although *she* was open.
- IV(b) **Emma** did not pass the ball to Janie, although *she* should have.
-
- V(a) Medvedev will cede the presidency to **Putin** because *he* is more popular.
- V(b) **Medvedev** will cede the presidency to Putin because *he* is less popular.
-

Table 1: Sample twin sentences. The target pronoun in each sentence is italicized, and its antecedent is boldfaced.

GLUE: General Language Understanding Evaluation (and there is also SuperGLUE...)

The General Language Understanding Evaluation (GLUE) benchmark is a collection of resources for training, evaluating, and analyzing natural language understanding systems. GLUE consists of:

- A benchmark of nine sentence- or sentence-pair language understanding tasks built on established existing datasets and selected to cover a diverse range of dataset sizes, text genres, and degrees of difficulty,
- A diagnostic dataset designed to evaluate and analyze model performance with respect to a wide range of linguistic phenomena found in natural language, and
- A public leaderboard for tracking performance on the benchmark and a dashboard for visualizing the performance of models on the diagnostic set.

The format of the GLUE benchmark is model-agnostic, so any system capable of processing sentence and sentence pairs and producing corresponding predictions is eligible to participate. The benchmark tasks are selected so as to favor models that share information across tasks using parameter sharing or other transfer learning techniques. The ultimate goal of GLUE is to drive research in the development of general and robust natural language understanding systems.

GLUE Tasks

Task Acronym	Task Name	Dataset for task	Expected output	Evaluation criteria
CoLA	Corpus of Linguistic Acceptability	Each example is a sequence of words annotated with whether it is a grammatical English sentence.		Matthews correlation coefficient
SST-2	Stanford Sentiment Treebank	Movie reviews and human annotations of their sentiment. They use only single sentence-level labels	Positive/Negative label for each review	Accuracy.
MRPC	Microsoft Research Paraphrase Corpus	Contains sentence pairs with human annotations for whether the sentences in the pair are semantically equivalent.	Predict if pair of sentences are semantically equivalent.	Accuracy and F1 score.
QQP	Quora Question Pairs	Collection of question pairs.	Predict if question pairs are semantically equivalent.	Accuracy and F1 score.
STS-B	Semantic Textual Similarity Benchmark	Collection of sentence pairs. Each is annotated with a similarity score from 1 to 5.	Predict the similarity scores for each pair.	Pearson and Spearman correlation coefficients.
MNLI (Matched and Mismatched)	Multi-Genre Natural Language Inference Corpus	Collection of sentence pairs with textual entailment annotations. It contains a premise sentence and a hypothesis sentence.	Predict whether premise entails the hypothesis(entailment), contradicts the hypothesis(contradiction), or neither(neutral).	Accuracy.
QNLI	Question-answering NLI, Question paragraph pairs from SQuAD.	Collection of question-paragraph pairs. Converted into a sentence pair classification, each question and sentence in corresponding context. Pairs with low lexical overlap are filtered out. Modifies	Determine whether the context sentence contains answer to the question.	Accuracy.
RTE	Recognizing Textual Entailment	Combines a series of annual textual entailment challenges.	Predict entailment/not_entailment. Datasets having three-class split, neutral and contradiction are collapsed into not_entailment.	Accuracy.
WNLI	Winograd NLI, derived from Winograd Schema Challenge	Reading comprehension task, system must read a sentence with a pronoun and select the referent of that pronoun from a list of choices. It is converted to a sentence pair classification, by replacing the ambiguous pronoun with each possible referent.	Predict if the sentence with the pronoun substituted is entailed by the original sentence.	Accuracy. Each example is evaluated separately.

Super GLUE Tasks

Task Acronym	Task Name	Dataset for task	Expected output	Evaluation criteria
BoolQ	Boolean Questions	A QA task consisting of a short passage and a yes/no question about the same.	Predict the correct answer of the question.	Accuracy.
CB	Commitment Bank	A collection of short texts in which at least one sentence contains an embedded clause. Each example consists of a premise containing an embedded clause and corresponding hypothesis is the extraction of that clause,	Three-class textual entailment.	Accuracy and F1 score.
COPA	Choice of Plausible Alternatives	Causal reasoning task.	A premise sentence is given, system must determine either the cause or the effect of the premise from two possible choices.	Accuracy.
MultiRC	Multi-Sentence Reading Comprehension	A QA task, each example consisting of a context paragraph, a question about that paragraph, and a list of possible answers.	Predict which answers are true and which are false. Each question can have multiple possible correct answers.	F1 over all answer-options (F1a) and exact match of each question's set of answers (EM).
ReCoRD	Reading Comprehension with Commonsense Reasoning Dataset	Multiple-choice QA task. Each example consists of a news article and a Cloze-style question about the article in which one entity is masked out.	Predict the masked out entity from a given list of possible entities in the provided passage, where the same entity may be expressed using multiple different surface forms, all of which are considered correct.	Max (over all mentions) token-level F1 and exact match (EM).
RTE	Recognizing Textual Entailment	Same as GLUE.		Accuracy.
WiC	Word-in-Context	Word sense disambiguation task cast as binary classification of sentence pairs. Given two text snippets and a polysemous word that appears in both sentences.	Determine whether the word is used with the same sense in both sentences.	Accuracy.
WSC	Winograd Schema Challenge	Almost same as GLUE.		Accuracy.



- MMLU (Massive Multitask Language Understanding) is a benchmark designed to evaluate the broad knowledge and reasoning abilities of AI language models. It consists of a wide variety of multiple-choice questions covering 57 diverse topics, including math, history, medicine, law, and more. The questions range in difficulty and are structured to simulate real-world standardized tests and academic exams.
- Average Human Baseline: Typically, non-expert human performance on MMLU benchmarks averages around 50-60%. GPT-4: Performance scores for GPT-4 and similar models reach up to 86% or higher in some tests, illustrating how these models now frequently exceed non-expert human performance and match or approach that of domain specialists in some cases.
- Reasoning vs. Memorization: MMLU aims to difference between genuine reasoning ability and the model's capacity to recall memorized information, pushing researchers to enhance models' logical thinking capabilities.
- See also: <https://en.wikipedia.org/wiki/MMLU>

Task	Tested Concepts	Supercategory
Abstract Algebra	Groups, rings, fields, vector spaces, ...	STEM
Anatomy	Central nervous system, circulatory system, ...	STEM
Astronomy	Solar system, galaxies, asteroids, ...	STEM
Business Ethics	Corporate responsibility, stakeholders, regulation, ...	Other
Clinical Knowledge	Spot diagnosis, joints, abdominal examination, ...	Other
College Biology	Cellular structure, molecular biology, ecology, ...	STEM
College Chemistry	Analytical, organic, inorganic, physical, ...	STEM
College Computer Science	Algorithms, systems, graphs, recursion, ...	STEM
College Mathematics	Differential equations, real analysis, combinatorics, ...	STEM
College Medicine	Introductory biochemistry, sociology, reasoning, ...	Other
College Physics	Electromagnetism, thermodynamics, special relativity, ...	STEM
Computer Security	Cryptography, malware, side channels, fuzzing, ...	STEM
Conceptual Physics	Newton's laws, rotational motion, gravity, sound, ...	STEM
Econometrics	Volatility, long-run relationships, forecasting, ...	Social Sciences
Electrical Engineering	Circuits, power systems, electrical drives, ...	STEM
Elementary Mathematics	Word problems, multiplication, remainders, rounding, ...	STEM
Formal Logic	Propositions, predicate logic, first-order logic, ...	Humanities
Global Facts	Extreme poverty, literacy rates, life expectancy, ...	Other
High School Biology	Natural selection, heredity, cell cycle, Krebs cycle, ...	STEM
High School Chemistry	Chemical reactions, ions, acids and bases, ...	STEM
High School Computer Science	Arrays, conditionals, iteration, inheritance, ...	STEM
High School European History	Renaissance, reformation, industrialization, ...	Humanities
High School Geography	Population migration, rural land-use, urban processes, ...	Social Sciences
High School Gov't and Politics	Branches of government, civil liberties, political ideologies, ...	Social Sciences
High School Macroeconomics	Economic indicators, national income, international trade, ...	Social Sciences
High School Mathematics	Pre-algebra, algebra, trigonometry, calculus, ...	STEM
High School Microeconomics	Supply and demand, imperfect competition, market failure, ...	Social Sciences
High School Physics	Kinematics, energy, torque, fluid pressure, ...	STEM
High School Psychology	Behavior, personality, emotions, learning, ...	Social Sciences
High School Statistics	Random variables, sampling distributions, chi-square tests, ...	STEM
High School US History	Civil War, the Great Depression, The Great Society, ...	Humanities
High School World History	Ottoman empire, economic imperialism, World War I, ...	Humanities
Human Aging	Senescence, dementia, longevity, personality changes, ...	Other
Human Sexuality	Pregnancy, sexual differentiation, sexual orientation, ...	Social Sciences
International Law	Human rights, sovereignty, law of the sea, use of force, ...	Humanities
Jurisprudence	Natural law, classical legal positivism, legal realism, ...	Humanities
Logical Fallacies	No true Scotsman, base rate fallacy, composition fallacy, ...	Humanities
Machine Learning	SVMs, VC dimension, deep learning architectures, ...	STEM
Management	Organizing, communication, organizational structure, ...	Other
Marketing	Segmentation, pricing, market research, ...	Other
Medical Genetics	Genes and cancer, common chromosome disorders, ...	Other
Miscellaneous	Agriculture, Fermi estimation, pop culture, ...	Other
Moral Disputes	Freedom of speech, addiction, the death penalty, ...	Humanities
Moral Scenarios	Detecting physical violence, stealing, externalities, ...	Humanities
Nutrition	Metabolism, water-soluble vitamins, diabetes, ...	Other
Philosophy	Skepticism, phronesis, skepticism, Singer's Drowning Child, ...	Humanities
Prehistory	Neanderthals, Mesoamerica, extinction, stone tools, ...	Humanities
Professional Accounting	Auditing, reporting, regulation, valuation, ...	Other
Professional Law	Torts, criminal law, contracts, property, evidence, ...	Humanities
Professional Medicine	Diagnosis, pharmacotherapy, disease prevention, ...	Other
Professional Psychology	Diagnosis, biology and behavior, lifespan development, ...	Social Sciences
Public Relations	Media theory, crisis management, intelligence gathering, ...	Social Sciences
Security Studies	Environmental security, terrorism, weapons of mass destruction, ...	Social Sciences
Sociology	Socialization, cities and community, inequality and wealth, ...	Social Sciences
US Foreign Policy	Soft power, Cold War foreign policy, isolationism, ...	Social Sciences
Virology	Epidemiology, coronaviruses, retroviruses, herpesviruses, ...	Other
World Religions	Judaism, Christianity, Islam, Buddhism, Jainism, ...	Humanities

Table 2: Summary of all 57 tasks.

Beyond MMLU ...

While MMLU (Massive Multitask Language Understanding) is a very challenging and diverse benchmark covering 57 academic and professional tasks, there are a few datasets and benchmarks that are considered even more challenging in specific ways, particularly in testing reasoning, robustness, truthfulness, or multi-step problem solving.

Dataset	Focus	What makes it more challenging than MMLU
BBH	Broad reasoning tasks	Curated for tasks LLMs struggle with
ARC-Challenge	Science reasoning	Harder questions from standardized exams
TruthfulQA	Truthfulness, adversarial QA	Designed to cause hallucinations
GSM8K / MATH	Multi-step math reasoning	Requires structured, chain-of-thought
HellaSwag	Commonsense reasoning	Includes adversarial distractors

Measuring Factuality



Maastricht University

Teaching note: A model can sound fluent and still be factually wrong — none of the metrics so far actually check this.

Chatbots ...

- There are several aspects relevant to validate the quality of a chatbot.
- We could measure basic linguistic phenomena such as correct inflections, -syntax, -semantics, etc.
- Some have to do with the “natural” flow of a dialog and whether a chatbot “sounds like another human being”.
- Others have to do with creativity, and factuality (aka correctness).
- Most chatbots using generative AI “sound” like humans, but factuality is often a problem.

Let’s look at a number of possible evaluation metrics ...



The Ragas Framework combines all of the above:

Ragas is a framework that helps you evaluate your Retrieval Augmented Generation (RAG) pipelines. RAG denotes a class of LLM applications that use external data to augment the LLM's context. There are existing tools and frameworks that help you build these pipelines but evaluating it and quantifying your pipeline performance can be hard. This is where Ragas (RAG Assessment) comes in.

Hint

Ground truth: Einstein was born in 1879 in Germany.

High answer correctness: In 1879, Einstein was born in Germany.

Low answer correctness: Einstein was born in Spain in 1879.

Also includes code
to generate part of
the data set.

<https://docs.ragas.io/>

```
from datasets import Dataset
from ragas.metrics import faithfulness, answer_correctness
from ragas import evaluate

data_samples = {
    'question': ['When was the first super bowl?', 'Who won the most super bowl?'],
    'answer': ['The first superbowl was held on Jan 15, 1967', 'The most super bowl won by the Pittsburgh Steelers'],
    'ground_truth': ['The first superbowl was held on January 15, 1967', 'The Pittsburgh Steelers won the most super bowl']
}

dataset = Dataset.from_dict(data_samples)
score = evaluate(dataset, metrics=[answer_correctness])
score.to_pandas()
```



Evaluation Methods for different NLP Tasks

Task	Type	Methods
Classification (NER, Sentiment, POS, ...)	Balanced	Accuracy
	Un-balanced	Precision, Recall, F-1, 11 Points PR, ROC
Abstracting		ROUGE, METEOR, BLEU
Machine Translation		BLEU, METEOR
Dialogue / Q&A		SQuAD, MMLU
Quality of a Language Model		Perplexity
Language “Understanding” and Linguistic Acceptability (correct style, grammar, ...)		Winograd, GLUE, Super GLUE, MMLU
Clustering & Topic Modeling*		Purity, Rand Index, Pointwise Mutual Information (PMI), Silhouette Index

One measurement, a few or all of them?

Despite their simplicity, language models are increasingly functioning as the foundation for almost all language technologies from question answering to summarization. But their immense capabilities and risks are not well understood. Holistic Evaluation of Language Models (HELM) is a living benchmark that aims to improve the transparency of language models.

1. **Broad coverage and recognition of incompleteness.** We define a taxonomy over the scenarios we would ideally like to evaluate, select scenarios and metrics to cover the space and make explicit what is missing.



2. **Multi-metric measurement.** Rather than focus on isolated metrics such as accuracy, we simultaneously measure multiple metrics (e.g., accuracy, robustness, calibration, efficiency) for each scenario, allowing analysis of tradeoffs.

	Metrics						
	Accuracy	Calibration	Robustness	Fairness	Bias	Toxicity	Efficiency
RAFT	✓	✓	✓	✓	✓	✓	✓
IMDB	✓	✓	✓	✓	✓	✓	✓
Natural Questions	✓	✓	✓	✓	✓	✓	✓
QuAC	✓	✓	✓	✓	✓	✓	✓
XSUM	✓				✓	✓	✓

3. **Standardization.** We evaluate all the models that we have access to on the same scenarios with the same adaptation strategy (e.g., prompting), allowing for controlled comparisons. Thanks to all the companies for providing API access to the limited-access and closed models and [Together](#) for providing the infrastructure to run the open models.

AI21labs

ANTHROPIC

BigScience

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Google

Meta

Microsoft

NVIDIA

OpenAI



Yandex

TOGETHER

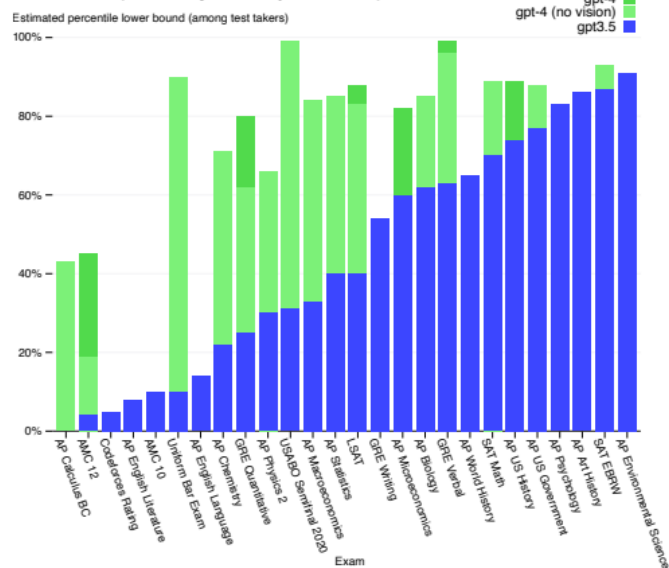
<https://venturebeat.com/ai/stanford-debuts-first-ai-benchmark-to-help-understand-llms/>
<https://crfm.stanford.edu/helm/latest/>

But Nothing Beats “Beating Humans”



Or Acing Human Exams!

Exam results (ordered by GPT-3.5 performance)



Source: GPT-4 Technical Report. OpenAI, March 2023. <https://arxiv.org/pdf/2303.08774.pdf>

Exam	GPT-4	GPT-4 (no vision)	GPT-3.5
Uniform Bar Exam (MBE+MEE+MPT)	298 / 400 (~90th)	298 / 400 (~90th)	213 / 400 (~10th)
LSAT	163 (~88th)	161 (~83rd)	149 (~40th)
SAT Evidence-Based Reading & Writing	710 / 800 (~93rd)	710 / 800 (~93rd)	670 / 800 (~87th)
SAT Math	700 / 800 (~89th)	690 / 800 (~89th)	590 / 800 (~70th)
Graduate Record Examination (GRE) Quantitative	163 / 170 (~80th)	157 / 170 (~62nd)	147 / 170 (~25th)
Graduate Record Examination (GRE) Verbal	169 / 170 (~99th)	165 / 170 (~96th)	154 / 170 (~63rd)
Graduate Record Examination (GRE) Writing	4 / 6 (~54th)	4 / 6 (~54th)	4 / 6 (~54th)
USABO Semifinal Exam 2020	87 / 150 (99th - 100th)	87 / 150 (99th - 100th)	43 / 150 (31st - 33rd)
USNCO Local Section Exam 2022	36 / 60	38 / 60	24 / 60
Medical Knowledge Self-Assessment Program	75 %	75 %	53 %
Codeforces Rating	392 (below 5th)	392 (below 5th)	260 (below 5th)
AP Art History	5 (86th - 100th)	5 (86th - 100th)	5 (86th - 100th)
AP Biology	5 (85th - 100th)	5 (85th - 100th)	4 (62nd - 85th)
AP Calculus BC	4 (43rd - 59th)	4 (43rd - 59th)	1 (0th - 7th)
AP Chemistry	4 (71st - 88th)	4 (71st - 88th)	2 (22nd - 46th)
AP English Language and Composition	2 (14th - 44th)	2 (14th - 44th)	2 (14th - 44th)
AP English Literature and Composition	2 (8th - 22nd)	2 (8th - 22nd)	2 (8th - 22nd)
AP Environmental Science	5 (91st - 100th)	5 (91st - 100th)	5 (91st - 100th)
AP Macroeconomics	5 (84th - 100th)	5 (84th - 100th)	2 (33rd - 48th)
AP Microeconomics	5 (82nd - 100th)	4 (60th - 82nd)	4 (60th - 82nd)
AP Physics 2	4 (66th - 84th)	4 (66th - 84th)	3 (30th - 66th)
AP Psychology	5 (83rd - 100th)	5 (83rd - 100th)	5 (83rd - 100th)
AP Statistics	5 (85th - 100th)	5 (85th - 100th)	3 (40th - 63rd)
AP US Government	5 (88th - 100th)	5 (88th - 100th)	4 (77th - 88th)
AP US History	5 (89th - 100th)	4 (74th - 89th)	4 (74th - 89th)
AP World History	4 (65th - 87th)	4 (65th - 87th)	4 (65th - 87th)
AMC 10 ³	30 / 150 (6th - 12th)	36 / 150 (10th - 19th)	36 / 150 (10th - 19th)
AMC 12 ³	60 / 150 (45th - 66th)	48 / 150 (19th - 40th)	30 / 150 (4th - 8th)
Introductory Sommelier (theory knowledge)	92 %	92 %	80 %
Certified Sommelier (theory knowledge)	86 %	86 %	58 %
Advanced Sommelier (theory knowledge)	77 %	77 %	46 %
Leetcode (easy)	31 / 41	31 / 41	12 / 41
Leetcode (medium)	21 / 80	21 / 80	8 / 80
Leetcode (hard)	3 / 45	3 / 45	0 / 45

Table 1. GPT performance on academic and professional exams. In each case, we simulate the conditions and scoring of the real exam. We report GPT-4's final score graded according to exam-specific rubrics, as well as the percentile of test-takers achieving GPT-4's score.



Questions?



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Teaching note: Recap: accuracy $\rightarrow$ precision/recall/F1 $\rightarrow$ perplexity $\rightarrow$ ROUGE/BLEU $\rightarrow$ human/benchmark evaluation — pick the metric that matches what “good” means for your task.

Exercise 1

Suppose an medical search engine evaluates a corpus of $N = 100,000$ documents. Exactly 1,000 documents are relevant, while the remaining 99,000 irrelevant. A baseline classification model retrieves 500 documents in total, of which 400 are actually relevant.

- Construct the confusion matrix by calculating the exact counts for True Positives (tp), False Positives (fp), False Negatives (fn), and True Negatives (tn).
- Compute the model's **Accuracy**, **Precision**, and **Recall**.
- Compare these results to the "Snoogle.com" dummy baseline described in the lecture (a system that always retrieves 0 documents). Why does raw accuracy fail as an informative metric in unbalanced NLP and IR settings?
- In legal and medical domains, "lack of precision is cost, while lack of recall is risk". Using the generalized F_β formula from slide 16:

$$\beta^2 P + R$$
$$\text{with } F_\beta = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- Should a legal compliance team select $\beta = 0.5$ or $\beta = 2.0$? Justify your choice based on how β weights precision versus recall.

Exercise 2

Consider the following reference and candidate sentences produced by a machine translation/summarization model:

- **Reference (R):** *"the doctor confirmed the patient was cured"*
- **Candidate 1 (C_1):** *"the patient was cured"*
- **Candidate 2 (C_2):** *"the doctor confirmed the patient was not cured"*
- Calculate unigram (p_1) and bigram (p_2) precision for both C_1 and C_2 using BLEU clipping rules against the reference.
- Explain how the **Brevity Penalty (BP)** in BLEU penalizes candidate C_1 . Why is a brevity penalty mathematically necessary when evaluating translation via precision-based n-gram overlap?
- Compute ROUGE-1 recall for C_2 against the reference. Notice that C_2 obtains a near-perfect ROUGE/BLEU score despite containing a critical factual contradiction ("not cured" vs. "cured"). Based on Tang et al. (2023), why can standard n-gram overlap metrics provide dangerous overconfidence in high-stakes domains (e.g., medicine or law)?
- How do semantic metrics like **COMET** or **BERTScore** resolve the structural blind spots of BLEU and ROUGE?

Exercise 3

Language models are evaluated across different levels of the data–information–knowledge continuum, from token probability modeling to pragmatic reasoning.

- What is **Perplexity** in terms of test set probability and entropy? If Model A achieves a perplexity of 16.5 on WikiText and Model B achieves 5.5, which model is a better predictor of the test distribution?
- Why do frontier chat models (such as GPT-4, Claude 3, and Gemini) report benchmark accuracies on MMLU, GSM8K, and HumanEval rather than publishing perplexity scores? Explain the impact of instruction tuning / RLHF on perplexity comparability.
- Consider the classic Winograd Schema pair:
 - *Sentence A: "The city councilmen refused the demonstrators a permit because they feared violence."*
 - *Sentence B: "The city councilmen refused the demonstrators a permit because they advocated violence."*
 - Why can standard syntactic parsing and lexical co-occurrence resolve neither pronoun reference without commonsense world knowledge? At which level of the continuum (Syntax, Semantics, Pragmatics, or Discourse) does this task reside?

Discussion Lecture: Quantitative Measurements



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Teaching note: Use today's metrics (precision/recall, perplexity, BLEU/ROUGE) as concrete ammunition — there's no single correct answer to these.

On the Surface Overlap Illusion: When High Scores Produce Dangerous Outputs

In both BLEU and ROUGE, evaluation is built around surface n-gram overlap with reference sentences. Consider a clinical summary where the reference states: 'The patient was cured', and an LLM outputs: 'The patient was not cured'.

That output achieves almost near-perfect ROUGE and BLEU scores, yet it is catastrophically false.

If our automated metrics can yield high scores on factually inverted or lethal claims, what are they actually measuring?

When we optimize models against lexical overlap, are we evaluating genuine language quality, or are we simply encouraging models to gamble on superficial similarity?

What is the role of error analysis?

On the Disappearance of Perplexity: Next-Token Likelihood vs. Real-World Intelligence

Perplexity is mathematically the gold standard for measuring how well a language model predicts text distributions—lower perplexity directly equals higher test-set likelihood.

Yet the leading frontier LLMs (like GPT-4, Claude, and Gemini) avoid publishing perplexity entirely, shifting entirely to standardized exams and benchmarks like MMLU, GSM8K, and HumanEval.

Did we discard perplexity simply because RLHF and instruction tuning alter the pure likelihood objective, or does minimizing perplexity hit a philosophical limit where knowing the most probable next word ceases to correlate with true reasoning and factuality?

If a model achieves near-zero perplexity on a corpus full of human falsehoods, is it an intelligent model or just a perfect parrot?

On the Winograd Schema: Can Co-occurrence Ever Solve Commonsense Pragmatics?

- In the classic Winograd example—'The city councilmen refused the demonstrators a permit because they feared/advocated violence'—the syntax is identical in both sentences, yet changing one verb flips the referent of 'they'.
- Slide 34 highlights that this decision lives at the pragmatic 'knowledge level,' where the necessary facts exist neither in the text nor in its lexicon, but in shared commonsense reality.
- Can an LLM trained exclusively on textual token co-occurrence ever genuinely resolve pragmatics, or is it merely executing high-dimensional pattern matching that simulates understanding without possessing the conceptual model of how the real world operates?

References

- More information on most of the metrics discussed in this lecture can be found in: Introduction to Information Retrieval. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze. Cambridge University Press, 2008. <http://informationretrieval.org>
- Metrics on machine translation, dialogs and deeper language understanding can be found in: Speech and Language Processing (Dan Jurafsky and James H. Martin). Prentice Hall, 2021. <https://web.stanford.edu/~jurafsky/slp3/>
- And the various other references used in this lecture.